Inference and assumptions

Last time: Activity

Data on 116 sparrows. Want to make a 98% confidence interval.

```
qt(..., df = ..., lower.tail=FALSE)
```

> What do I fill in to calculate the critical value t^*?

qt(0.01 , 114 , lower.tail= FALSE)

n-2

Activity

Data on 116 sparrows. Want to make a 98% confidence interval.

```
qt(0.01, df = 114, lower.tail=FALSE)
```

```
## [1] 2.359504
```

Activity

```
             Estimate Std. Error
(Intercept)  8.75465    1.39601
Weight       1.31341    0.09756
```

$t^* = 2.36$

What is my 98% confidence interval?

$\hat{\beta}_1 \pm t^* SE_{\hat{\beta}_1}$

$1.313 \pm 2.36(0.0976) = (1.08, 1.54)$

Activity

```
              Estimate Std. Error
(Intercept)   8.75465    1.39601
Weight        1.31341    0.09756
```

$t^* = 2.36$

$$1.31 \pm 2.36(0.098) = (1.08, 1.54)$$

> How do I interpret this confidence interval?

We are 98% confident that a one-gram increase in sparrow weight is associated with an increase in wing length by between 1.08 and 1.54 mm

Activity

```
             Estimate Std. Error
(Intercept)  8.75465    1.39601
Weight       1.31341    0.09756
```

$t^* = 2.36$

$$1.31 \pm 2.36(0.098) = (1.08, 1.54)$$

> Your friend claims that there is a 2% chance that the true slope β_1 is greater than 1.54 or less than 1.08. Is your friend correct?

Activity

$$1.31 \pm 2.36(0.098) = (1.08, 1.54)$$

> Your friend claims that there is a 2% chance that the true slope β_1 is greater than 1.54 or less than 1.08. Is your friend correct?

No. The true slope β_1 is a fixed (though unknown) parameter. There is no randomness involved in β_1, we just don't know what it is.

This means that for any specific interval we calculate -- e.g., (1.08, 1.54) -- β_1 is either *in the interval* or it is *not in the interval*. There is no probability involved!

Probability only becomes involved when we imagine repeating the process *many times*.

Linear regression assumptions

- **Shape assumption:** the shape of the regression model is at least approximately correct

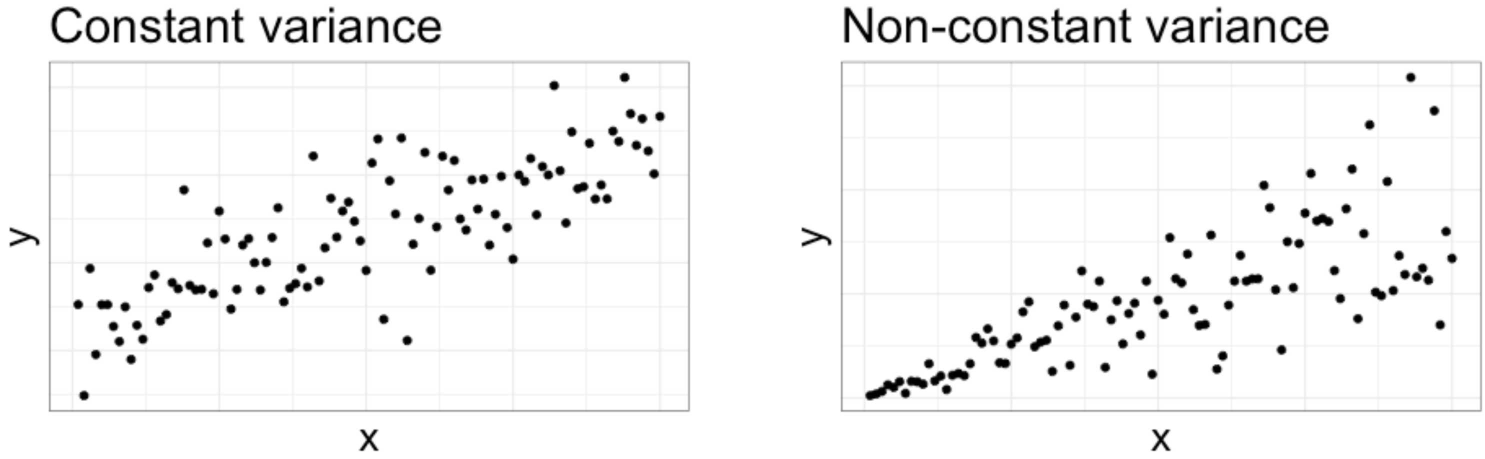
To do *inference* (confidence intervals and hypothesis tests) we need some other assumptions:

- Constant variance
- Normality (with zero mean)
- Randomness
- Independence

Constant variance

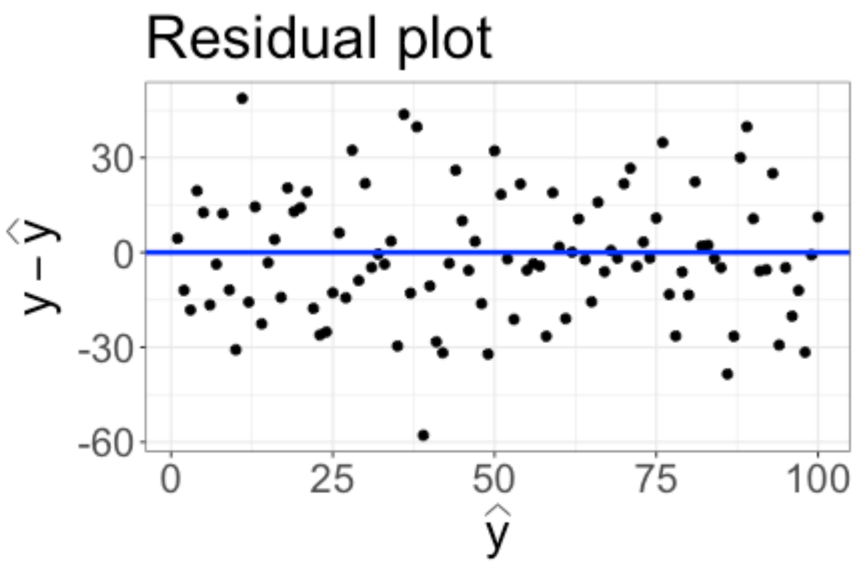
$$y = \beta_0 + \beta_1 x + \varepsilon$$

Assumption: Variance of the noise ε is the same for all values of the predictor x

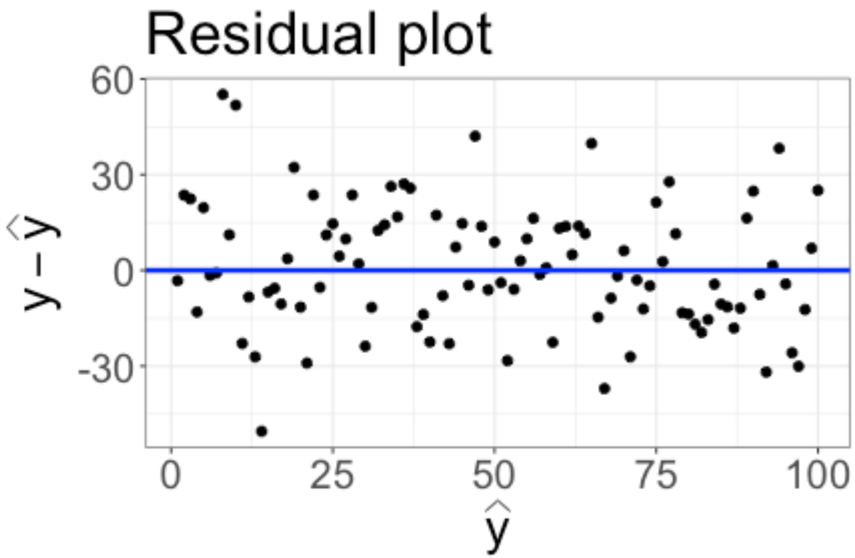


Assessing shape and constant variance: residual plots

Residual plot: plot residuals $y - \hat{y}$ on vertical axis, and predicted values $\hat{y}$ on horizontal axis. Add a horizontal line at 0.



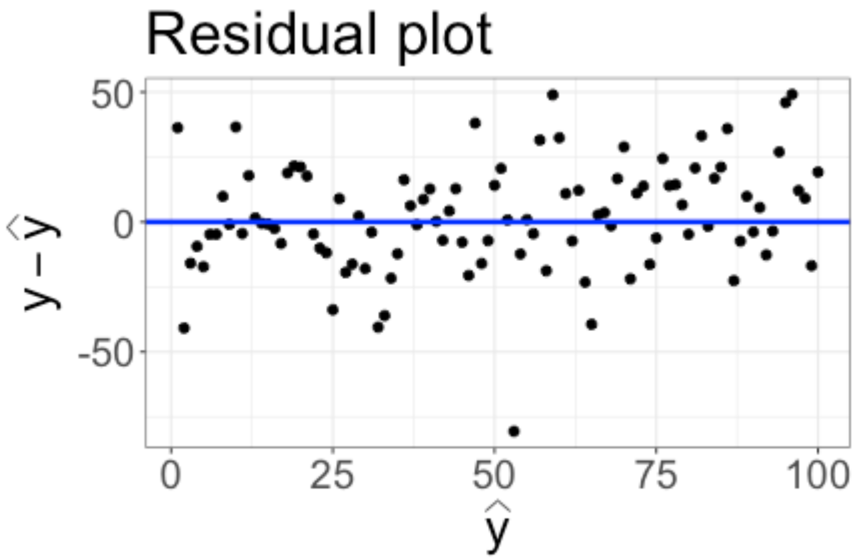
Residual plots



Shape assumption is reasonable if:

- residuals appear to be scattered randomly above and below 0
- no clear patterns in the residuals

Residual plots



Constant variance assumption is reasonable if:

- the band of residuals has relatively constant width

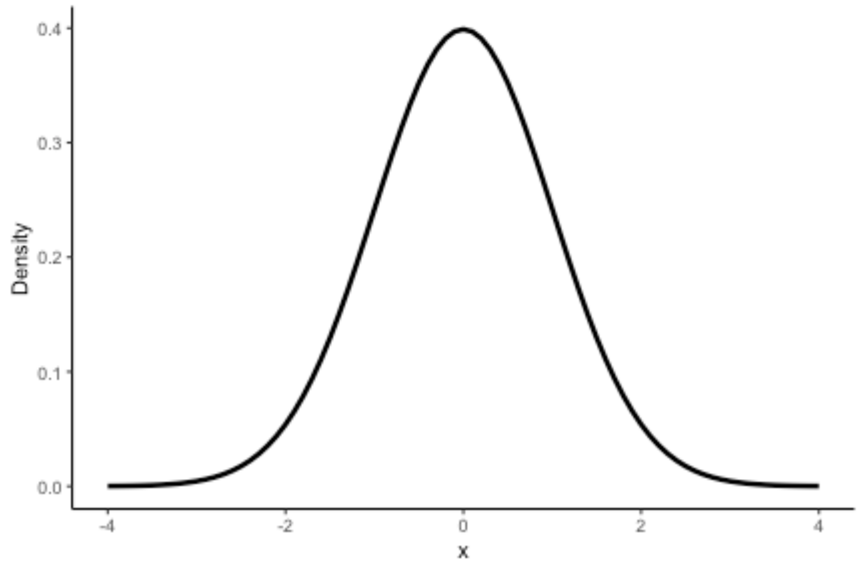
Normality (with zero mean)

Linear regression model:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Assumption: The noise term ε follows a <u>normal</u> distribution, with mean 0

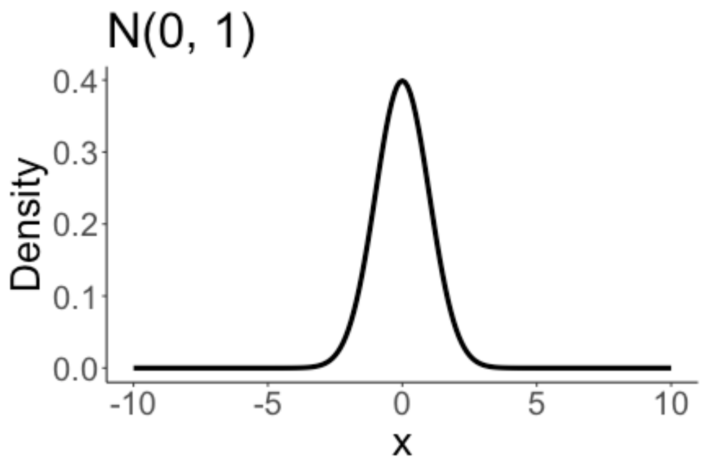
Example of a normal distribution:



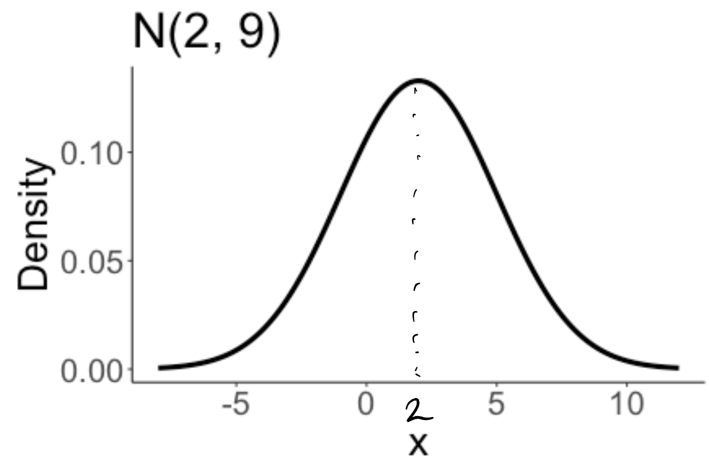
Normal distributions

Parameterized by mean μ and variance σ^2:

$$N(\mu, \sigma^2)$$



(standard normal)



mean = 2

variance = 9

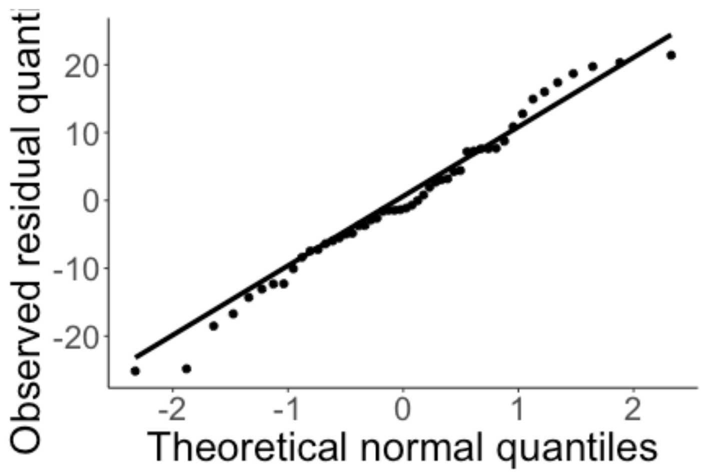
$\Rightarrow$ SD = 3

Assessing normality

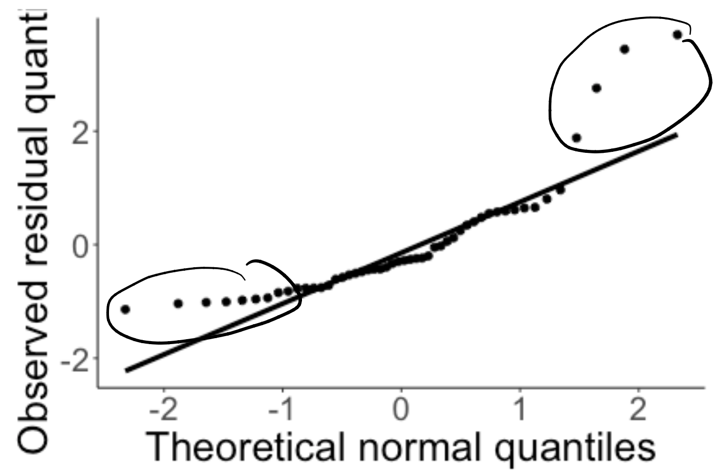
Quantile-quantile (QQ) plots: quantiles of residuals on y-axis, quantiles of theoretical normal distribution on x-axis.

If residuals are normal, plot should look approximately linear.

Residuals look normal

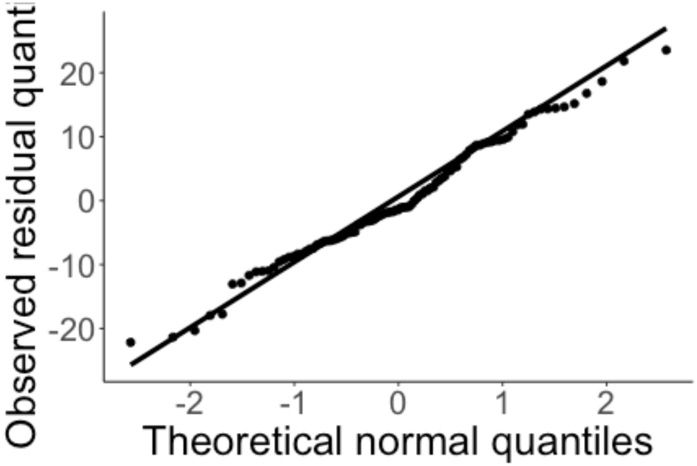


Residuals do not look normal

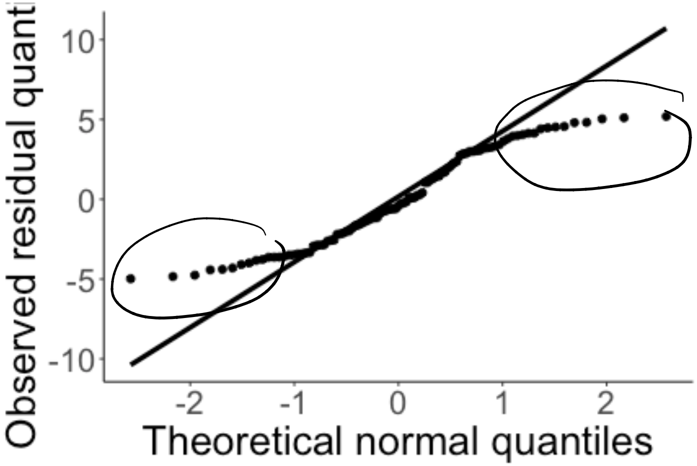


QQ plot examples

Residuals look normal



Residuals do not look normal



Formal assumptions for inference

Assumptions about the relationship between variables:

- Shape
- Constant variance
- Normality (with zero mean)

Assumptions about how the data were collected:

- Independence
- Randomness

Independence

Assumption: Observations are independent -- one point falling above or below the line has no influence on the location of another point

Independence can't be checked with plots. We need to think about how the data were collected.

Independence?

Scenario: You collect a sample of 116 sparrows from Kent island. For each sparrow, you record their wing length and weight. Your data has 116 rows (one for each sparrow), and two columns.

Is it reasonable to assume the observations are independent?

(A) Yes

(B) No

The 116 sparrows are independent – one sparrow's information doesn't affect another sparrow

Independence?

Scenario: You go to Kent island, and find a sparrow. For 116 consecutive days, you record its weight and wing length. Your data has 116 rows, and two columns.

Is it reasonable to assume the observations are independent?

(A) Yes

(B) No

we have multiple observations
on the same sparrow

Randomness

Assumption: The data are obtained using a random process, such as a random sample or randomized experiment

Randomness assumption is reasonable

We collect data on a random sample of 116 sparrows from Kent island. For each sparrow, we record weight and wing length.

Randomness assumption is not reasonable

We collect data on the weight and wing length of sparrows from Kent island. To collect the data, I deliberately choose sparrows that are very heavy, with very short wings.

Randomness

Assumption: The data are obtained using a random process, such as a random sample or randomized experiment

Randomness assumption is reasonable

We collect data on a random sample of 116 sparrows from Kent island. For each sparrow, we record weight and wing length.

Conclusions from the estimated regression line can be generalized to a population of interest.

Randomness assumption is not reasonable

We collect data on the weight and wing length of sparrows from Kent island. To collect the data, I deliberately choose sparrows that are very heavy, with very short wings.

Conclusions from the estimated regression line do not generalize.

Randomness

Assumption: The data are obtained using a random process, such as a random sample or randomized experiment

- Assumption about data collection process
- Necessary for defining the scope of inference
- Can't be checked with plots

Putting it all together

For a quantitative response y and a single quantitative predictor x , the simple linear regression model is

shape assumption

$$y = \overbrace{\beta_0 + \beta_1 x}^{\text{shape}} + \varepsilon$$

where $\varepsilon \sim \underbrace{N(0, \sigma_\varepsilon^2)}_{\text{normality}}$, and the noise ε are independent from one another.

normality
constant variance

independence

When we fit the model with a sample $(x_1, y_1), \ldots, (x_n, y_n)$, we also assume the data are generated by a random process that allows us to generalize to the population of interest.

Summary: checking assumptions for inference

- Shape and constant variance: residual plot
- normality: QQ plot
- independence and randomness: think about data generation and collection

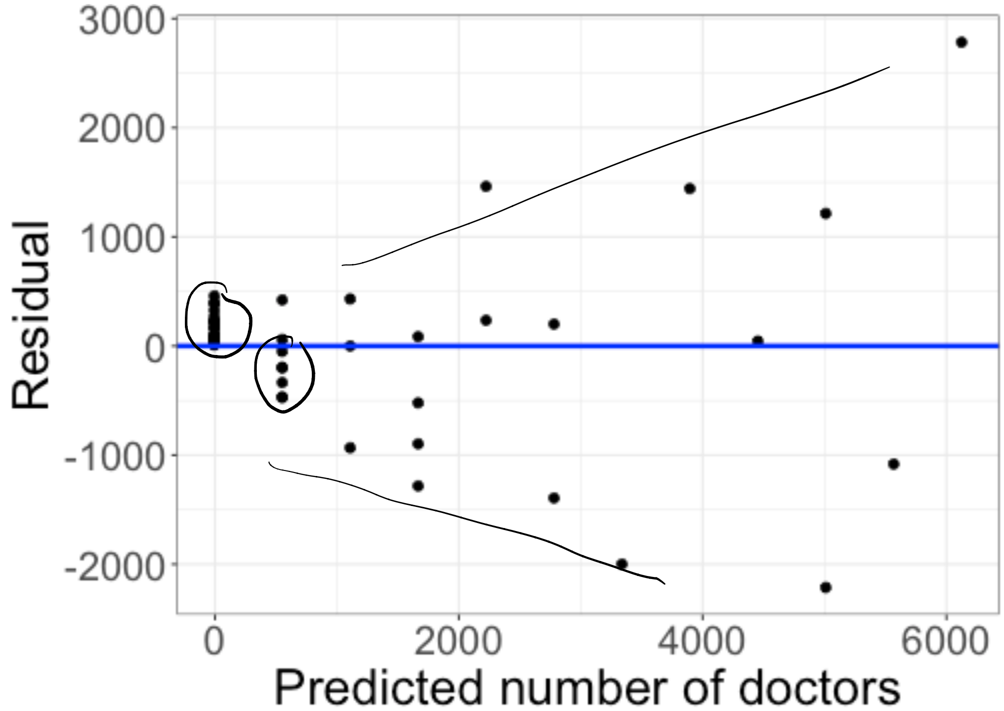
Class activity

Data on doctors and hospitals from a random sample of 53 US counties:

- `County`: the name and state of the county
- `MDs`: the number of medical doctors in the county
- `Hospitals`: number of hospitals in the county
- `Beds`: the number of beds in the hospital

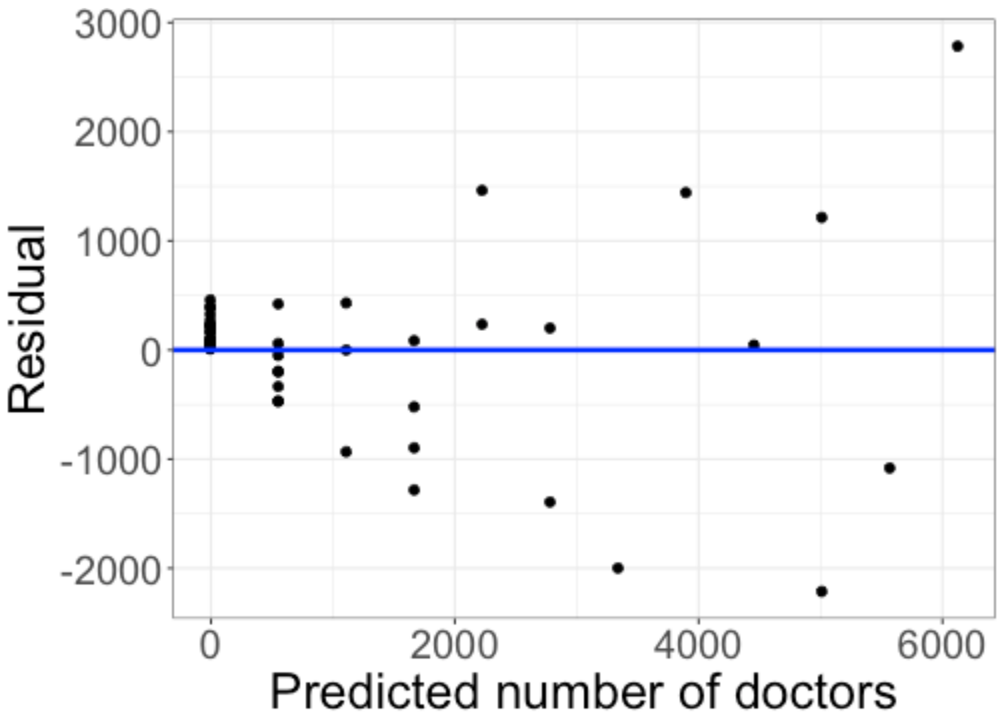
https://sta112-s26.github.io/class_activities/ca_14.html

Class activity



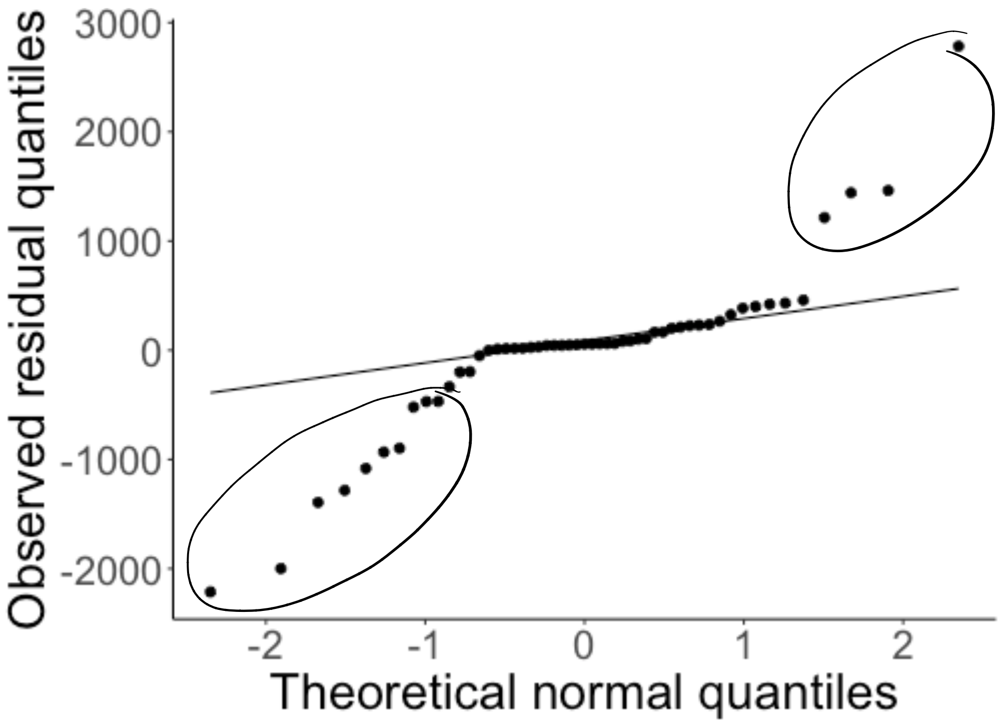
Do the shape and constant variance assumptions look reasonable?

Class activity



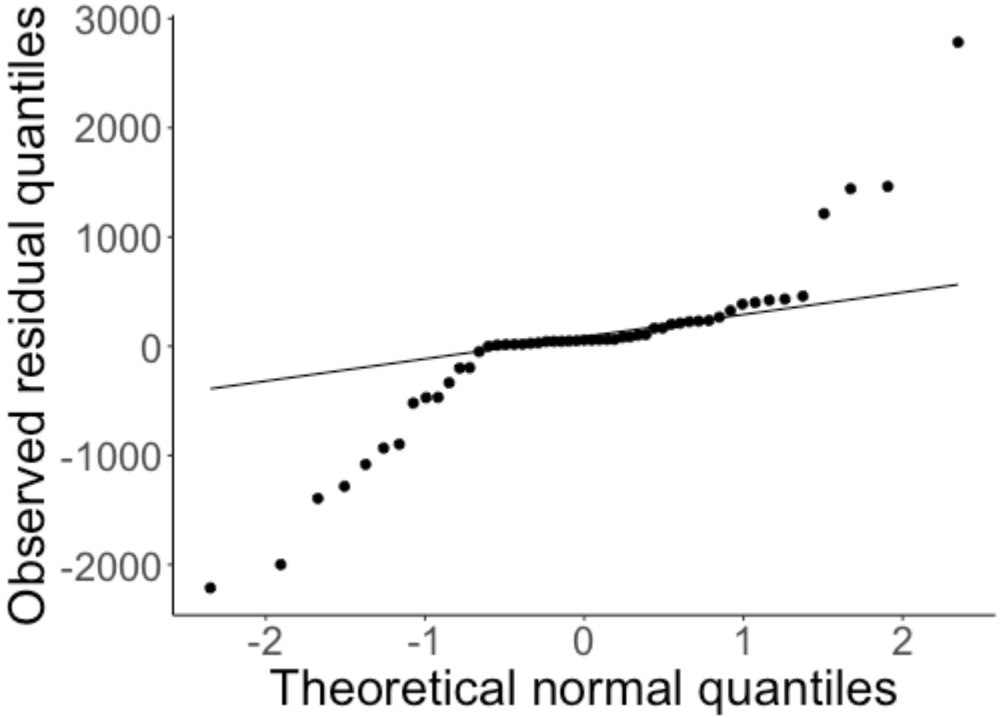
- Shape may be slightly violated
- Constant variance is definitely violated

Class activity



> Does the normality assumption look reasonable?

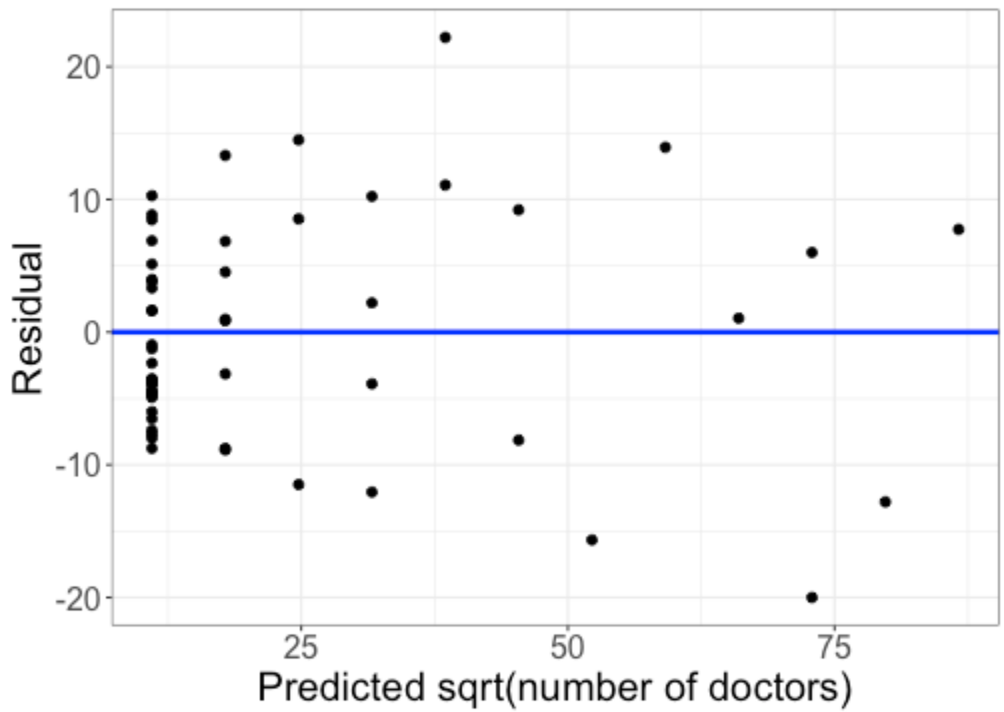
Class activity



The normality assumption is definitely violated.

Transformation

Trying a square root transformation of the response:

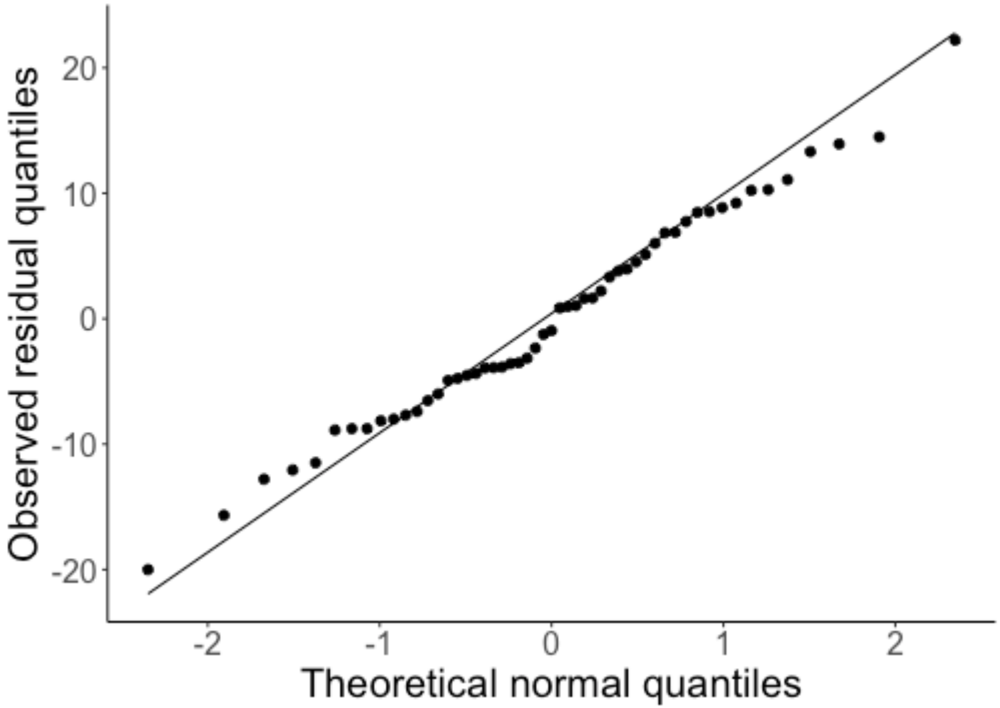


> How do the shape and constant variance assumptions look now?

much better!

Transformation

QQ plot with the transformed data:



How does the normality assumption look now? much better!

Class activity

> Do the independence and randomness assumptions seem reasonable for the `CountyHealth` data?

Class activity

> Do the independence and randomness assumptions seem reasonable for the `CountyHealth` data?

Yes, both seem reasonable.

- Independence: reasonable to assume the number of hospitals/doctors for one county doesn't affect the number of hospitals/doctors for another county
- Randomness: we are told the data come from a random sample